

Tony Nguyen

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Education

M.S. in Artificial Intelligence, Machine Learning Northeastern University	<i>Expected May 2028</i> Seattle, WA
B.A. in Computer Science, Data Science & Economics	<i>May 2026</i>
B.B.A. in Business Administration, Finance Gonzaga University President's Honor List Data Science TA <i>Summa Cum Laude</i> GPA: 3.96	Spokane, WA

Technical Proficiencies

Coding	Python (pandas, NumPy, scikit-learn, PyTorch, spaCy, FastAPI, pytest, Flask, Streamlit), SQL (MySQL, BigQuery SQL), C++, Java, Git, Linux.
Data Engineering	Tableau, Power BI, Microsoft Fabric, Salesforce CRM, Docker, Google Cloud (BigQuery, Vertex AI, Cloud Storage, Cloud Run, Compute Engine), AWS (EC2, S3, Lambda), ETL.
AI / LLM	RAG pipelines, LangChain, CrewAI, Semantic Kernel, Hugging Face Transformers, MCP, AI agents, embeddings.

Research & Industry Experience

AI / Machine Learning Engineer – Research Assistant <i>Institute for Informatics & Applied Technology – Gonzaga University</i>	January 2025 – May 2026 Spokane, WA
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- Designed and built the **Wazuh MCP Server**, an open-source Python server (**30+ endpoints**) enabling LLMs to query and analyze security events from a Wazuh SIEM platform through structured tool calls.
- Designed a benchmarking study comparing **skill integration** across frontier models against a ground-truth attack chain, demonstrating that structured domain knowledge closes the performance gap between smaller and larger models.
- Benchmarked **three LLMs** on **60+ CVE reports** using a three-stage RAG pipeline with specialized agents for retrieval, analysis, and remediation recommendation, focusing on insightfulness and verifiability.
- Developed and delivered a workshop on **RAG systems and LLM evaluation** to **30+ participants**, covering classical metrics, agent-based evaluation, and responsible AI practices.

Data Engineer Intern <i>KPIM Vietnam</i>	June 2025 – September 2025 Hanoi, Vietnam
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- Designed and maintained **10+ data pipelines** using SQL, Python, and Microsoft Fabric for **two Big Data projects**, enforcing schema validation, data quality checks, and optimizations that made querying **50% faster** than the client's internal system.
- Built reusable tooling and documentation to streamline troubleshooting, standardize workflows, and provide reliable, self-service financial and operational data for the client's six hotel properties.

Data Engineer – Innovation Intern <i>KONE Elevators & Escalators</i>	June 2024 – August 2024 Singapore
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- Built a centralized **Customer 360** by integrating **20 years** of data from **five countries**; aligned models to enterprise architecture and recommended schema optimizations to improve consistency and join performance.
- Developed a **Tableau** dashboard using SAP, Salesforce, and Excel data; created models and calculations to monitor customer behavior, service KPIs, and revenue in real time.
- Engineered ETL pipelines on **AWS S3** for ingestion, storage, and validation; automated integrity checks and partitioning to **reduce data retrieval latency by 40%** and improve reliability.

Data Science & Machine Learning Intern <i>TecAlliance Vietnam</i>	June 2023 – August 2023 Ho Chi Minh City, Vietnam
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- Engineered Python algorithms to parse data from **100+ wiring-diagram SVGs**, creating structured datasets for computer vision models (YOLO) and advanced ML pipelines.
- Deployed a **Neo4j graph database** for relationships extracted from automotive schematics, increasing **query speed by 80%** over the client's existing database and enabling deeper insights.
- Led weekly global meetings of nearly **15 team members** to unify data engineering efforts, consolidate findings, and manage timelines across multiple time zones.

Project

Minecraft AI Companion Applied AI Developer Senior Capstone	August 2025 – May 2026
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- Built an open-source Minecraft mod using a **two-model architecture** (local LLM for real-time dialogue and cloud model for retrieval-augmented queries), designing the context pipeline, tool-calling interface, and content guardrails for safe in-game interactions.

Mortality Risk Simulator Project Lead	Spring 2025
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- Developed a neural-network mortality-risk prediction pipeline using demographic, vitals, and medical-history features to support accurate and fair underwriting review.